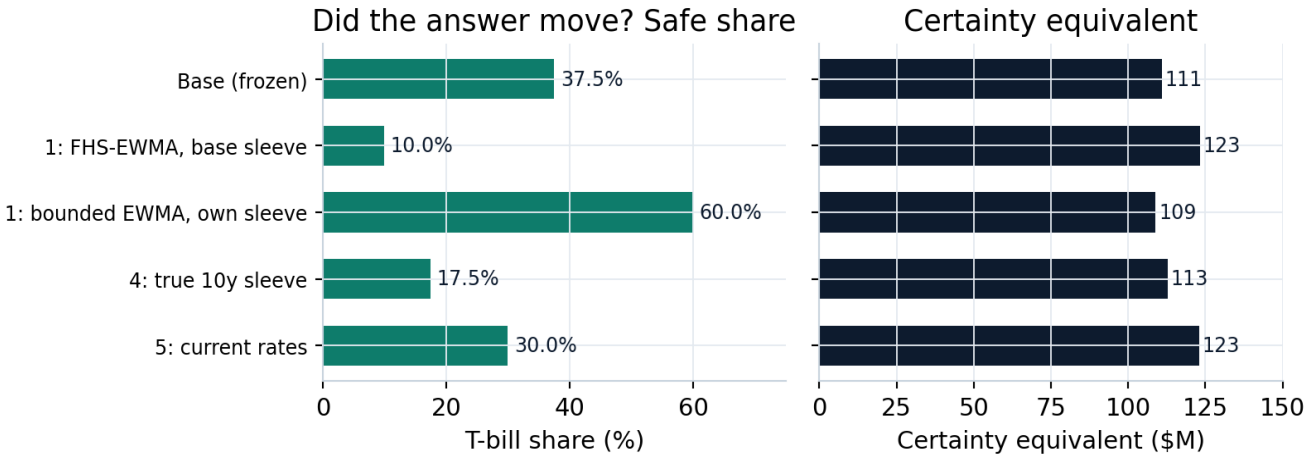
Open points 1-6: what the fixes changed

Six open points from the model review, worked through on the frozen 10-year setup (T-bills + TIPS model, $250k/month for 120 months, 40% any-point loss cap, 15% terminal loss limit). Every variant below is a labeled sensitivity run under the same validation discipline: the portfolio is picked on validation draws and re-judged on independent final draws, with a one-step-safer walk-back. Nothing here changes an engine default.

The base answer before the fixes: **37.5% T-bills**, certainty equivalent **$110.8M**, all 120 payments funded.

The one-line answer per fix

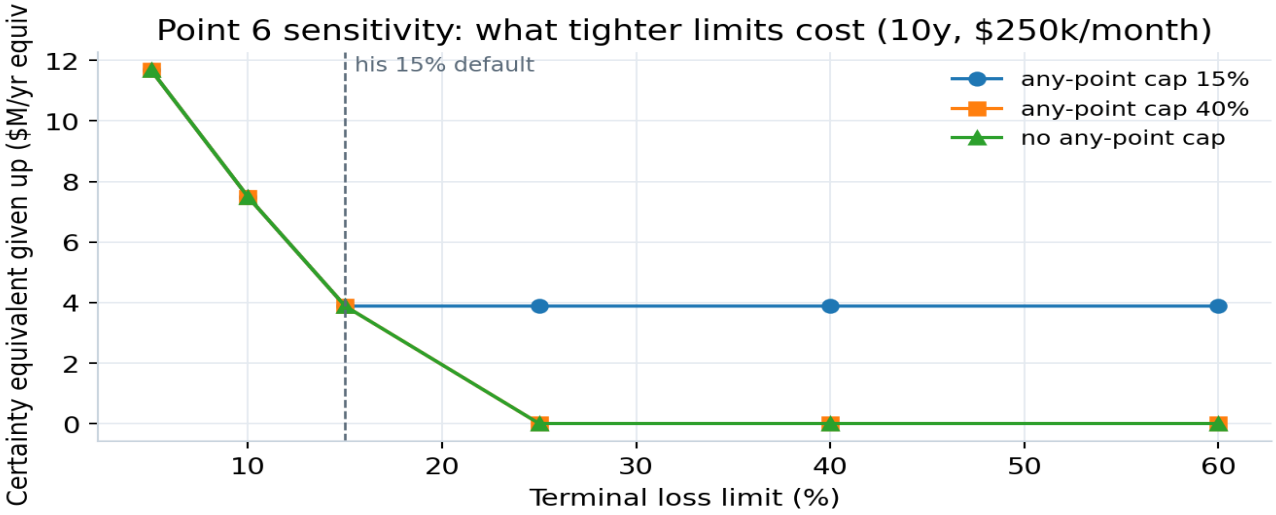
Fix	What was done	Did the chosen portfolio move?
1 market model too well-behaved	Bootstrap/regime variants (FHS-EWMA, bounded EWMA mixture) fitted and judged	Yes, both directions: 10% and 60% bills (from 37.5%)
2 model-miss radius uncalibrated	Calibrated from refit instability + radius grid	No portfolio change possible: any real radius breaks the budgets
3 everything nominal	Full real-terms variant: CPI-deflated draws, indexed payouts, TIPS as the real safe asset	No mix passes the 15% real limit at all - the nominal safety is partly an illusion
4 1y-proportion screens	True 10-year optimization for the risky sleeve	Yes: 17.5% bills, more IEF; CE +$1.9M
5 rerun at current rates	Safe-asset carry at 5 Nov 2026 quotes (bill 4.17%, 10y 4.84%, TIPS real 2.60%)	Yes: 30% bills, CE $123.1M (+$12.2M)
6 constraint tightness	Sensitivity grid only, nothing changed	n/a - grid below



Certainty equivalent (CE) is the sure yearly-dollar answer that scores the same as the risky plan after funding every payment - higher is better, and differences of $1-2M are small against the $110.8M base.

Point 6 - the constraint sensitivity (nothing changed)

How much upside tighter or looser limits give up, around the 40% any-point cap and 15% terminal loss limit defaults. Two clean results. First, at his 40% default the **any-point cap never binds**: cap 40% or no cap at all give the identical portfolio and certainty equivalent at every terminal limit in the grid. Second, the **terminal loss limit is the one with teeth**:



At his 15% default the plan gives up **$3.88M** against the loosest setting; at 10% it gives up $7.49M; at 5%, $11.69M. Loosening from 15% to 25% recovers the whole $3.88M; past 25% there is nothing more to recover - the curve is flat. One nuance on the cap: at his 40% default it never binds, but a tight 15% cap starts to bind once the terminal limit is loosened to 25% or more (it holds the safe share at 37.5% where the plan would otherwise drop to 10% bills).

Fix 1 - a less well-behaved market model

The fitted iid mixture is calm: squared-return autocorrelation about 0.00 against 0.32 in history, so it misses vol clustering. Two bootstrap/regime variants were fitted and run through the same pipeline: **FHS-EWMA** (filtered historical simulation, today's vol regime) and a **bounded EWMA mixture**.

FHS-EWMA sees a calm current regime (equity vol 9.6% vs 18% fitted) and, on the frozen base sleeve, the safe share **falls to 10% bills** (CE $123.3M). The bounded EWMA mixture pushes the other way once its own sleeve is re-solved: **60% bills** (CE $108.9M). Every cross-judgment passes (each model's answer stays valid under the other's draws).

Honest caveat: the FHS-EWMA run as wired is a current-vol-regime law - its lower safe share is a regime statement ("vol is calm right now"), not just "the same world with fatter tails". The answer is genuinely model-dependent, which is the point of the open point: the mix moves by 27 points of bills either way depending on which law of motion is true.

Fix 4 - true long-horizon optimization

Instead of screening 1-year sleeve proportions onto 10 years, the risky sleeve was solved directly on the 10-year problem (n=100 in 8.6s, n=200 in 76s; n=400 is out of sandbox reach and labeled as such). The 10-year solve wants much more 7-10yr Treasuries in the sleeve (56% of risky vs 40% at 1y).

Portfolio	T-bills	IEF 7-10y	US stocks	Gold	TIPS	SG prop.
Base: 1y sleeve, fitted means	37.5%	21.2%	17.5%	10.6%	3.2%	10.0%
Fix 4: true 10y sleeve	17.5%	40.9%	18.1%	13.5%	0.0%	10.0%
Fix 5: current rates	30.0%	14.9%	24.8%	20.3%	0.0%	10.0%

Fix 4 moves the answer to **17.5% bills** and lifts the certainty equivalent to **$112.7M** (+$1.9M vs base) - the 1-year screen was leaving a little on the table by under-using the bond fund.

Fix 5 - current rates as a second default view

The fitted window carries bills at 1.81%; the market quote on 5 Nov 2026 is 4.17% (10y 4.84%, 10y TIPS real 2.60%, FRED H.15). With safe-asset carry set to current quotes (vols, correlations and risky means unchanged - a labeled simplification), the answer moves to **30% bills** and the certainty equivalent rises to **$123.1M** (+$12.2M). TIPS drop to zero at a 2.6% real quote against 4.8% nominal on the 10y. This is a second default view worth adopting next to the fitted-means view.

Fix 3 - real terms: the uncomfortable one

Everything was re-run in today's dollars: every scenario path deflated by a bootstrapped inflation path (US CPI 1947-2026, mean 3.5%/yr, 10-year regime spread 1.8%-6.5%), payouts inflation-indexed, TIPS modeled as the real safe asset at the live 2.60% real yield. Two modeling traps were caught on the way: treating TIPS as a nominal-quote override, and deflating TIPS by the simulated inflation path - both quietly turn TIPS back into nominal bonds exactly in the high-inflation states where they should protect.

No static mix passes the 15% real terminal loss limit at 10 years. Not the base sleeve, not a re-solved sleeve, not 90% TIPS (the most the fixed 10% property allows), at fitted means or at current rates. The reason is structural: 10-year inflation-regime uncertainty puts about ±20% of real dispersion into every nominal asset, and the only real asset (TIPS) still carries ~6% real-yield vol. Judged in real terms, today's base portfolio is worth a certainty equivalent of **$71.6M** - against $110.8M nominal - and it breaches both real loss tails. In plain words: the 15% limit reads as protection in dollars-of-the-day, and a large part of that protection is an inflation illusion.

What a real limit can be: scanning budgets with the judge unchanged, the tightest pair any candidate passes is about **45% any-point / 40% terminal** - for 90% TIPS + 10% property at current rates, real CE $80.5M. If a real limit is wanted, it has to be looser than the nominal one, or the horizon has to shorten.

Fix 2 - the model-miss radius

The buffer radius was calibrated from **refit instability** as asked: bootstrap refits of the fitting window, distances between refit and base scenario sets in the declared path geometry. Calibrated radius: **2.55** (q90; the existing past-empirical 52-week method gives 0.41 on its own horizon). The current manual radius is **2e-5** - four to five orders of magnitude smaller.

The radius grid shows why: the current 2e-5 already spends $40k of the $50k smallest per-payment budget on buffer alone. At 2.5e-5 the answer is unchanged (37.5% bills); at 3e-5 every mix fails. A radius calibrated to how much the fitted model actually moves on refitting is unaffordable under the declared budgets by orders of magnitude. So the model-miss buffer is cosmetic by construction - making it real is a budget decision (looser limits or a different distance geometry), not a calibration decision.

Caveats that apply to everything above

Private research on owner-accepted Yahoo data; parametric scenarios, not a backtest. Inflation is bootstrapped independently of asset returns (labeled simplification). The judge's tail bounds are Monte Carlo diagnostics over 4 batches, not economic coverage. Current-quotes carry is a snapshot of 5 Nov 2026. No engine default was changed; every variant is a labeled study under studies/fixes/.